ModelIt!

Student Activity Pack

What Makes Wind Blow?

The Invisible Force That Moves the Air

Grade 2 | NGSS 2-ESS2-3 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Can you see wind? How do you know when it is windy outside?**

2. **What do you think makes the air start moving?**

3. **Draw a picture showing a windy day — how can we tell it is windy in your picture?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sun Heating | Air Movement | Wind Strength

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Hot Sunny Day

Sun Heating set to high

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Cool Cloudy Day

Sun Heating set to low

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Beach Breeze

Sun Heating set to high over land, low over water

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Surface Type — What the ground is made of — dark pavement heats up faster than grass or water, creating temperature differences

Time of Day — How the sun's position changes heating — midday sun heats more than morning or evening sun

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

You cannot see wind, but you can feel it — what makes the air start moving? Use evidence from your simulation to support your answers.

1 If you cannot see wind, how do you know it is there?

2 Why do you think it is often breezier near the ocean or a lake?

8 STEM Challenge: Build a Wind Detector

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YOUR ENGINEERING MISSION

Design and build a simple tool that can detect wind and show which direction it is blowing.

THE SCENARIO

A weather station at your school needs a tool to measure wind. Your team has been asked to build a wind detector that shows when wind is blowing and which direction it comes from. Test it outside!

Engineering Design Thinking

1. WHAT LIGHTWEIGHT MATERIALS WILL YOU USE THAT CAN MOVE IN THE WIND — STREAMERS, PAPER, OR FABRIC?
2. HOW WILL YOUR DETECTOR BE ATTACHED SO IT DOES NOT BLOW AWAY?
3. HOW WILL YOU TEST YOUR DETECTOR — BY A FAN FIRST, THEN OUTSIDE?
4. HOW CAN YOU IMPROVE YOUR DETECTOR AFTER THE FIRST TEST?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Meteorologists study weather patterns, including wind, to predict storms, temperature changes, and weather conditions. They use tools like weather vanes and anemometers every day. They earn $50,000–$95,000/year.